
PNT Lab Internship – Final Report

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Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

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BCA-ARTIFICIAL INTELLIGENCE(AI)

JAIN (Deemed-to-be University)

Under the Guidance of

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Supported by

Geospatial Intelligence and Applications Laboratory,

IIT Tirupati Navavishkar I-Hub Foundation

WEEK 1 REPORT

1. Basic Information

Field	Details
Intern Name	YUNES ABDULGHANI MOHAMMED GHALEB
Enrollment No.	23BCAR0626
Institution	JAIN (Deemed-to-be University)
Faculty Mentor / PI	Dr. K. Suneetha, Dr. Zion Ramdinthara
Week No.	Week 1 (July 7 - July 13, 2025)
Reporting Date	July 13, 2025

2. Project Details

Project Title:

Project Title: Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

Focuc Area:

GNSS Security, NavIC, Spoofing Attacks, Machine Learning, Artificial Intelligence

Objective for this Week:

The primary objective of Week 1 was to build a **strong conceptual foundation** in Global Navigation Satellite Systems (GNSS) with a specific focus on:

- Understanding how GNSS systems operate at a fundamental level
- Studying the architecture and characteristics of the **NavIC (Navigation with Indian Constellation)** system
- Identifying **security vulnerabilities** in GNSS signals
- Gaining an in-depth understanding of **GNSS spoofing attacks**, their mechanisms, and real-world consequences.

The primary objective of the first week was orientation into the STAR PNT Internship program, understanding the fundamentals of GNSS and NavIC systems, identifying the security challenges associated with GNSS spoofing attacks, conducting an initial literature survey, and finalizing the project problem statement along with a high-level methodological roadmap.

3. Background Study and Initial Orientation

At the beginning of the internship, an orientation session was conducted to understand the **problem statement**, project expectations, and overall workflow.

The project focuses on **GNSS spoofing detection using Artificial Intelligence**, particularly applied to the **NavIC system**. Since GNSS is a safety-critical technology used in aviation, defense, autonomous systems, and telecommunications, understanding its vulnerabilities is essential before attempting any AI-based solution.

During this week, emphasis was placed on:

- Navigation principles
- Satellite-based positioning systems
- Signal-level behavior
- Security challenges

4. Tasks Undertaken

During the first week, an orientation session was conducted to familiarize interns with the objectives, structure, and expectations of the STAR PNT Internship Program. This session provided clarity on Positioning, Navigation, and Timing (PNT) applications and emphasized the importance of research documentation through weekly reports.

A detailed study of GNSS fundamentals was carried out, covering the working principles of satellite-based navigation systems and their role in providing accurate positioning, navigation, and timing services. Special focus was given to India's NavIC system, its regional coverage, signal characteristics, and advantages over conventional GPS in regional applications.

An extensive literature review was performed on GNSS spoofing attacks, focusing on their operational mechanisms, real-world impact, and potential risks to civilian and defense applications. Various academic research papers and technical articles were analyzed to understand how spoofing differs from jamming and why it poses a more severe security threat.

Existing GNSS spoofing detection methods were reviewed, including signal power monitoring, correlation distortion analysis, and hardware-based solutions. Limitations of traditional techniques were identified, particularly their inability to adapt to evolving spoofing strategies and their reliance on specialized hardware.

Based on mentor discussions and feasibility analysis, the project focus was finalized to develop an AI-based GNSS spoofing detection framework using machine learning models trained on GNSS/NavIC signal parameters. Google Colab was selected as the primary development environment due to its computational support and ease of experimentation.

4. Results / Outputs

By the end of the first week, the project scope was clearly defined, focusing on intelligent detection and classification of GNSS spoofing attacks using machine learning techniques. A clear understanding of GNSS security challenges and the importance of AI-driven approaches was established.

An initial project workflow was prepared, outlining data collection, preprocessing, feature extraction, model training, and evaluation stages. The decision to use NavIC-related signal features was documented as a key innovation of the project.

Preliminary documentation, including the problem statement and research motivation, was completed in alignment with the STAR PNT Internship guidelines.

5. Challenges and Solutions

One of the primary challenges during the first week was the complexity of GNSS signal-level concepts, particularly for understanding spoofing behavior in real-world scenarios. This challenge was addressed through structured study of foundational resources and continuous guidance from faculty mentors.

Another challenge involved narrowing down the project scope to ensure both innovation and practical feasibility within the internship duration. This was resolved by focusing specifically on AI-based spoofing detection using signal parameters rather than hardware-intensive solutions.

No major managerial or coordination issues were encountered during this week.

6. Plan for Next Week

In the upcoming week, the focus will shift towards exploring available GNSS spoofing datasets, including both synthetic and real signal data. A detailed study of relevant signal parameters such as carrier-to-noise ratio (C/N_0), Doppler shift, and signal stability will be conducted.

Additionally, various machine learning techniques suitable for anomaly detection and classification will be reviewed to identify potential candidate models. Data preprocessing strategies, including normalization and labeling methods, will also be explored to prepare for model training in subsequent weeks.

7. Learning Summary

During the first week, significant understanding was gained regarding GNSS and NavIC system fundamentals, GNSS security threats, and the critical nature of spoofing attacks. Exposure to research literature enhanced analytical skills and provided insight into current gaps in GNSS spoofing detection research.

The week also contributed to the development of professional skills, including research documentation, technical reporting, and effective communication with academic mentors.

8. Verification

Intern Signature:



Date: July 13, 2025

Faculty Mentor Signature:



Date: July 13, 2025

WEEK .2 REPORT

1. Basic Information

Field	Details
Intern Name	YUNES ABDULGHANI MOHAMMED GHALEB
Enrollment No.	23BCAR0626
Institution	JAIN (Deemed-to-be University)
Faculty Mentor / PI	Dr. K. Suneetha, Dr. Zion Ramdinthara
Week No.	Week 2 (July 13 - July 20, 2025)
Reporting Date	July 20, 2025

2. Project Details

Project Title:

Project Title: Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

Focus Area:

GNSS Spoofing Detection, NavIC Signal Analysis, Machine Learning, Data Preprocessing, Feature Engineering

Objective for this Week:

The objective of the second week was to design the complete conceptual system architecture for AI-based GNSS spoofing detection, study GNSS/NavIC signal parameters relevant to spoofing behavior, explore available datasets (synthetic and real), and initiate data preprocessing and feature understanding for machine learning model development.

3. Tasks Undertaken

During this week, a complete high-level system architecture was designed for the proposed GNSS spoofing detection framework. The architecture consisted of multiple stages, including data acquisition, signal parameter extraction, preprocessing, feature engineering, model training, and performance evaluation.

An in-depth study of GNSS and NavIC signal parameters was conducted to identify features that are sensitive to spoofing behavior. Particular attention was given to carrier-to-noise ratio (C/N_0), Doppler shift, signal power consistency, and temporal stability, as these parameters exhibit noticeable deviations under spoofing conditions.

Multiple GNSS spoofing datasets were explored during this week. Initially, synthetic datasets generated through simulation tools were reviewed to understand baseline spoofing characteristics. Subsequently, real-world GNSS/NavIC datasets were examined to assess noise, multipath effects, and realistic signal variations that are not captured in simulated data.

Google Colab was finalized as the primary experimental platform. The environment was configured with required Python libraries for data analysis and machine learning. Initial exploratory data analysis (EDA) was performed to understand data distributions, class imbalance, and feature correlations.

Additionally, preliminary discussions were held regarding the selection of suitable machine learning models. Traditional classifiers and advanced ensemble and deep learning approaches were evaluated conceptually, with the understanding that multiple models would be trained and compared in later stages to identify the most robust solution.

4. Results / Outputs

By the end of the second week, a clear and structured system architecture for AI-based GNSS spoofing detection was established. Key signal features relevant to spoofing detection were identified and documented.

A comparative understanding between synthetic and real GNSS datasets was achieved, highlighting the necessity of training and validating models on real-world data for improved generalization.

The Google Colab environment was fully configured, and initial data exploration results were documented as a foundation for model development in subsequent weeks.

Another challenge involved handling the complexity and volume of real-world GNSS datasets. This was mitigated by adopting structured preprocessing pipelines and leveraging Google Colab's computational resources.

5. Challenges and Solutions

One of the main challenges during this week was distinguishing meaningful spoofing indicators from normal signal fluctuations present in real GNSS data. This challenge was addressed through careful feature analysis and comparison between spoofed and non-spoofed samples.

6. Plan for Next Week

In the next week, focus will be placed on advanced data preprocessing, feature selection, and implementation of the first machine learning model. Model training will initially be performed on synthetic data, followed by gradual integration of real GNSS/NavIC datasets.

Performance metrics and evaluation strategies will also be defined to enable systematic comparison between multiple models in later stages.

7. Learning Summary

This week provided strong insights into GNSS signal behavior under spoofing conditions and reinforced the importance of real-world data in AI-based security research. Skills related to data analysis, system design, and experimental planning were significantly enhanced.

8. Verification

Intern Signature:



Date: July 20, 2025

Faculty Mentor Signature:



Date: July 20, 2025

WEEK .3 REPORT

1. Basic Information

Field	Details
Intern Name	YUNES ABDULGHANI MOHAMMED GHALEB
Enrollment No.	23BCAR0626
Institution	JAIN (Deemed-to-be University)
Faculty Mentor / PI	Dr. K. Suneetha, Dr. Zion Ramdinthara
Week No.	Week 3(July 20 - July 27, 2025)
Reporting Date	July 27, 2025

2. Project Details

Project Title:

Project Title: Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

Focus Area:

NavIC Signal Characteristics | GNSS Data Analysis | Feature Engineering | Machine Learning Model Design

Objective for this Week:

The objective of this week was to prepare the foundational dataset for model development by generating synthetic GNSS/NavIC data, performing preprocessing and feature extraction, conducting exploratory data analysis, and validating the feasibility of the machine learning pipeline through training a single baseline model.

3. Tasks Undertaken

During this week, the initial dataset preparation phase was carried out. Due to the limited availability of real-world GNSS spoofing datasets, synthetic GNSS/NavIC data was generated using Python in the Google Colab environment. This approach allowed controlled simulation of both normal and spoofed signal behaviors while maintaining reproducibility.

The synthetic dataset was designed to reflect realistic signal characteristics by varying parameters such as carrier-to-noise ratio (C/N_0), Doppler frequency shift, signal power variation, pseudorange stability, and temporal consistency. Spoofing scenarios were simulated by introducing structured anomalies in these parameters based on patterns reported in existing GNSS security literature.

Once generated, the dataset underwent extensive preprocessing. This included removing incomplete records, filtering corrupted measurements, and eliminating extreme outliers that could adversely affect model learning. Feature normalization and scaling techniques were applied to ensure numerical stability during training.

A feature extraction phase was then performed, focusing on signal parameters that are known to be sensitive to spoofing attacks. Exploratory Data Analysis (EDA) was conducted using statistical visualization techniques to compare normal and spoofed signal distributions, revealing noticeable behavioral differences between the two classes.

At this stage, only a single machine learning model was trained as a baseline to validate the overall workflow. The purpose of this step was to confirm that the generated data, extracted features, and preprocessing pipeline were sufficient for learning discriminative patterns, rather than achieving optimal classification performance.

4. Results / Outputs

By the end of this week, a fully prepared synthetic GNSS/NavIC dataset was available for experimentation. A validated preprocessing and feature extraction pipeline was established, ensuring consistency and reliability for subsequent modeling stages.

The baseline machine learning model successfully demonstrated the feasibility of distinguishing between normal and spoofed signal patterns, confirming that the selected features and data representation were suitable for spoofing detection tasks.

5. Challenges and Solutions

One major challenge was ensuring that the synthetic data realistically represented spoofing behavior. This challenge was addressed by grounding the data generation process in established GNSS spoofing literature and carefully tuning parameter variations.

Another challenge involved avoiding overfitting during early experimentation. This was mitigated by limiting the training to a single baseline model and postponing complex architectures to later stages.

6. Plan for Next Week (Revised)

In the upcoming week, the focus will shift toward enhancing the realism and complexity of the synthetic dataset. The initial dataset used in the current week was intentionally kept simple to validate the end-to-end machine learning pipeline using a single baseline model.

To better approximate real-world GNSS/NavIC signal behavior, a more advanced data generation approach will be implemented using custom Python scripts. This enhanced dataset will incorporate increased variability, temporal dependencies, and complex spoofing patterns.

Based on this improved dataset, multiple machine learning models will be trained and evaluated to analyze their comparative performance and robustness. Real-world GNSS/NavIC data acquisition and advanced model development will be addressed and documented in the later stages of the internship.

7. Learning Summary

This week strengthened practical understanding of GNSS signal behavior, spoofing characteristics, and data-driven security analysis. Significant experience was gained in dataset generation, preprocessing, and early-stage machine learning validation.

8. Verification

Intern Signature:



Date: July 27, 2025

Faculty Mentor Signature:



Date: July 27, 2025

WEEK 4 REPORT

1. Basic Information

Field	Details
Intern Name	YUNES ABDULGHANI MOHAMMED GHALEB
Enrollment No.	23BCAR0626
Institution	JAIN (Deemed-to-be University)
Faculty Mentor / PI	Dr. K. Suneetha, Dr. Zion Ramdinthara
Week No.	Week 4 (July 27 - August 3, 2025)
Reporting Date	August 3, 2025

2. Project Details

Project Title:

Project Title: Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

Focus Area:

GNSS Signal Analysis, Synthetic Data Modeling, Feature Engineering, Machine Learning
Model Development

Objective for this Week:

The objective of this week was to enhance the complexity and realism of the synthetic GNSS/NavIC dataset, perform advanced feature engineering, and begin training and evaluation of multiple machine learning models for spoofing detection.

3. Tasks Undertaken

During this week, a more advanced synthetic GNSS/NavIC data generation framework was implemented using Python in Google Colab. Unlike the initial simple dataset used in earlier weeks, this dataset incorporated higher variability in signal parameters to better resemble real-world signal behavior.

Complex spoofing scenarios were simulated by introducing gradual power takeover patterns, Doppler shift inconsistencies, temporal instability, and pseudorange anomalies. These enhancements allowed the dataset to reflect more realistic spoofing attack characteristics.

Advanced feature engineering was performed, focusing on parameters such as carrier-to-noise ratio (C/N_0), Doppler frequency drift, signal power fluctuation, temporal variance, and inter-satellite consistency metrics. Feature normalization techniques were applied to ensure stable model training.

During this week, multiple machine learning models were selected and implemented for comparative analysis. Initial versions of Random Forest and Deep Neural Network (DNN) models were trained using the enhanced synthetic dataset. Google Colab was used as the primary development and experimentation environment.

Baseline performance evaluation was conducted to identify strengths and weaknesses of each model before introducing real-world data in later stages.

5. Challenges and Solutions

One of the main challenges was ensuring that the synthetic data accurately reflected realistic spoofing behavior without becoming overly complex or unstable. This was addressed through incremental feature addition and continuous validation.

Another challenge involved tuning model hyperparameters to avoid overfitting on synthetic data. Cross-validation and regularization techniques were applied to mitigate this issue.

Computational constraints were managed effectively by leveraging Google Colab's GPU/CPU resources and optimizing code execution.

6. Plan for Next Week

In the next week, the focus will shift toward acquiring and preprocessing real-world GNSS/NavIC datasets. The trained models will be adapted and retrained using authentic data to evaluate their generalization capability.

Further model optimization and comparative evaluation will be conducted, and additional models may be introduced based on observed performance trends.

7. Learning Summary

This week strengthened practical understanding of synthetic data modeling, feature engineering, and comparative machine learning analysis for GNSS security applications. The importance of realistic data generation and systematic model evaluation was clearly established.

8. Verification

Intern Signature:



Date: August 3, 2025

Faculty Mentor Signature:



Date: August 3, 2025

WEEK .5 REPORT

1. Basic Information

Field	Details
Intern Name	YUNES ABDULGHANI MOHAMMED GHALEB
Enrollment No.	23BCAR0626
Institution	JAIN (Deemed-to-be University)
Faculty Mentor / PI	Dr. K. Suneetha, Dr. Zion Ramdinthara
Week No.	Week 5 (August3 - August10, 2025)
Reporting Date	August10, 2025

2. Project Details

Project Title:

Project Title: Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

Focus Area:

Realistic GNSS/NavIC Data Simulation | Multi-Model Training | Performance
Evaluation | Generalization Analysis

Objective for this Week:

The objective of this week was to evaluate the trained machine learning models on more complex and near-realistic GNSS/NavIC spoofing scenarios, compare multiple models systematically, and prepare the pipeline for integration with real-world NavIC datasets.

3. Tasks Undertaken

During this week, the synthetic GNSS/NavIC dataset developed in the previous week was further refined to include noise patterns, temporal drift, and partial spoofing behaviors that closely resemble real-world attack conditions.

Multiple machine learning models were trained and evaluated, including Random Forest, Support Vector Machine (SVM), and Deep Neural Network (DNN). Each model was trained using identical datasets to ensure fair comparison.

Model evaluation was performed using standard performance metrics such as accuracy, precision, recall, F1-score, and confusion matrices. Special attention was given to false positive and false negative rates, as these are critical in spoofing detection systems.

Generalization behavior was analyzed by testing models on unseen spoofing patterns that were not explicitly included during training. This step helped assess the robustness of each model.

All experiments were conducted using Python in Google Colab, ensuring reproducibility and efficient computation.

4. Results / Outputs

The comparative analysis revealed that ensemble-based models such as Random Forest provided strong baseline performance with stable detection accuracy, while deep learning models demonstrated better adaptability to complex spoofing patterns.

The evaluation showed that increasing data complexity significantly improved model robustness compared to earlier weeks that relied on simpler synthetic datasets.

The complete machine learning pipeline—from data generation and preprocessing to training and evaluation—was validated and finalized for transition to real-world NavIC signal data.

5. Challenges and Solutions

A major challenge was preventing overfitting due to the use of synthetic data. This was addressed by introducing variability, noise, and unseen spoofing patterns during testing.

Another challenge involved balancing model complexity and computational cost. Lightweight models were optimized for efficiency, while deeper models were tuned carefully to avoid excessive training time.

6. Plan for Next Week

In the next week, real-world GNSS/NavIC datasets will be introduced into the pipeline. The developed models will be retrained and validated using authentic signal data.

Further refinement of spoofing classification accuracy will be performed, and the most promising model will be selected for final deployment and reporting.

7. Learning Summary

This week provided deeper insight into multi-model comparison, robustness analysis, and the impact of data realism on spoofing detection accuracy. The transition from synthetic to real-world data was clearly structured and prepared.

8. Verification

Intern Signature:



Date: August 10, 2025

Faculty Mentor Signature:



Date: August 10, 2025

WEEK .6 REPORT

1. Basic Information

Field	Details
Intern Name	YUNES ABDULGHANI MOHAMMED GHALEB
Enrollment No.	23BCAR0626
Institution	JAIN (Deemed-to-be University)
Faculty Mentor / PI	Dr. K. Suneetha, Dr. Zion Ramdinthara
Week No.	Week 6 (August10 –August17, 2025)
Reporting Date	August17, 2025

2. Project Details

Project Title:

Project Title: Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

Focus Area:

Real GNSS/NavIC Data Analysis | Model Validation | Performance Characterization |
Robustness Evaluation

Objective for this Week:

The objective of this week was to transition from synthetic GNSS/NavIC datasets to real-world signal data, validate the trained machine learning models under realistic conditions, analyze performance degradation or improvement, and characterize the robustness of each model against genuine spoofing behaviors.

3. Tasks Undertaken

During this week, real GNSS/NavIC signal datasets were successfully acquired after an extensive search and validation process. These datasets captured realistic signal behavior, including environmental noise, temporal variations, receiver instability, and signal distortions that were not present in the previously generated synthetic data. Comprehensive preprocessing was applied to the real datasets to ensure compatibility with the existing machine learning pipeline. This included noise filtering, normalization, handling of missing or inconsistent records, and aligning all extracted features with the same feature space used in earlier weeks to enable fair model comparison. Multiple machine learning models were trained and evaluated using the real-world GNSS/NavIC datasets, incorporating both traditional and deep learning approaches. Their performance was systematically compared against results obtained earlier using synthetic data to assess generalization capability. A robustness analysis was conducted by exposing the trained models to mixed scenarios containing genuine GNSS/NavIC signals alongside subtle and partially spoofed patterns, allowing evaluation of model stability under realistic operating conditions. At this stage, no final model selection was enforced; instead, performance metrics were carefully documented to support informed model selection in subsequent weeks. The final models adopted for the system—Deep Neural Networks (DNN), Random Forest (RF), and XGBoost—are discussed in later sections of the report, as these models demonstrated superior accuracy and robustness. All experiments, training procedures, and validations were implemented using Python within the Google Colab environment to ensure reproducibility, scalability, and efficient utilization of computational resources.

4. Results / Outputs

The evaluation results showed a slight decrease in overall accuracy compared to synthetic datasets, which was expected due to the complexity of real-world signals. However, the trained models demonstrated strong generalization capability.

Among the evaluated models, the Random Forest model maintained stable performance, while the Deep Neural Network showed improved sensitivity to subtle spoofing characteristics.

The results confirmed that the feature engineering strategy developed in earlier weeks was effective for real-world GNSS/NavIC spoofing detection.

5. Challenges and Solutions

One major challenge was the increased noise and irregularity in real GNSS/NavIC data, which affected classification confidence. This was mitigated by adaptive normalization and robust preprocessing techniques.

Another challenge was the limited availability of labeled real spoofing data. This limitation was addressed by carefully combining real genuine signals with previously validated spoofing simulations.

6. Plan for Next Week

In the next week, extended testing will be performed to further refine model parameters. The best-performing model will be finalized for deployment, and detailed result analysis and documentation will be prepared for the final report and presentation.

7. Learning Summary

This week provided critical insights into the gap between synthetic and real GNSS/NavIC data. It highlighted the importance of robustness, realistic feature engineering, and careful evaluation when deploying AI models for security-sensitive applications.

8. Verification

Intern Signature:



Date: August 17, 2025

Faculty Mentor Signature:



Date: August 17, 2025

WEEK .7 REPORT

1. Basic Information

Field	Details
Intern Name	YUNES ABDULGHANI MOHAMMED GHALEB
Enrollment No.	23BCAR0626
Institution	JAIN (Deemed-to-be University)
Faculty Mentor / PI	Dr. K. Suneetha, Dr. Zion Ramdinthara
Week No.	Week 7 (August17 -24August, 2025)
Reporting Date	August 24, 2025

2. Project Details

Project Title:

Project Title: Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

Focus Area:

**Real GNSS/NavIC Data Analysis, Advanced Machine Learning Models, Model
Selection and Optimization**

Objective for this Week:

The primary objective of this week was to train and evaluate advanced machine learning models using real GNSS/NavIC datasets, apply sophisticated feature engineering and optimization techniques, compare model performance rigorously, and finalize the selection of the most accurate and robust models for GNSS spoofing detection.

3. Tasks Undertaken

During this week, real-world GNSS/NavIC datasets were fully integrated into the machine learning framework after extensive validation. These datasets exhibited realistic characteristics such as noise, temporal drift, satellite geometry variation, and subtle spoofing behaviors that closely resemble real attack scenarios.

Advanced feature engineering techniques were applied to enhance model discriminative capability. This included temporal feature aggregation, signal stability metrics, Doppler variance analysis, and statistical consistency checks across multiple satellites. Feature normalization and scaling strategies were refined to ensure stable convergence for all models.

Multiple advanced machine learning models were trained and evaluated, including Deep Neural Networks (DNN), Random Forest (RF), Support Vector Machines (SVM), and XGBoost classifiers. Hyperparameter optimization techniques such as grid search and adaptive learning rate tuning were applied to improve model performance.

A comprehensive comparative analysis was conducted using real GNSS/NavIC data, focusing on accuracy, precision, recall, F1-score, and robustness against subtle spoofing attacks. Based on this rigorous evaluation, three models—DNN, Random Forest, and XGBoost—were selected as the final models due to their superior performance, generalization capability, and robustness.

All experiments, training, and validation processes were conducted using Python in the Google Colab environment to ensure scalability, reproducibility, and efficient computational performance.

4. Results / Outputs

During this week, extensive experimental evaluations were conducted on real GNSS/NavIC datasets to assess the performance of advanced machine learning models under realistic operating conditions. Based on comprehensive quantitative analysis, three high-performing models—**Deep Neural Network (DNN)**, **Random Forest (RF)**, and **XGBoost**—were finalized due to their superior detection accuracy and robustness against spoofing attacks.

Model Performance Summary (Real Dataset Evaluation)

The evaluation results demonstrated clear performance differences among the selected models, as summarized below:

- **Deep Neural Network (DNN):**

- Accuracy: **94.1%**
- Precision: **94.6%**
- Recall: **93.7%**
- F1-Score: **94.1%**
- False Negatives: **417**

The DNN model exhibited strong overall performance and effective learning of nonlinear signal characteristics. However, its relatively higher false negative count indicates a slightly reduced sensitivity to subtle spoofing patterns compared to XGBoost.

- **Random Forest (RF):**

- Accuracy: **93.5%**
- Precision: **92.8%**
- Recall: **93.1%**
- F1-Score: **92.9%**
- False Negatives: **502**

Random Forest provided stable and consistent results with good generalization capability. Nevertheless, its higher false negative rate makes it less suitable for safety-critical GNSS security applications where missed detections can lead to severe consequences.

- **XGBoost (Selected Best Model):**

- Accuracy: **95.5%**
- Precision: **96.0%**
- Recall: **95.4%**
- F1-Score: **95.7%**
- False Negatives: **334**

XGBoost achieved the **highest overall accuracy** and the **lowest false negative count** among all evaluated models. Its gradient boosting architecture effectively captured complex nonlinear relationships within real GNSS/NavIC signals, making it highly reliable for spoofing detection. The high precision ensures minimal false alarms, while the strong recall highlights its sensitivity to subtle spoofing attacks.

Final Model Selection Outcome

Based on accuracy, robustness, and false negative minimization, **XGBoost was identified as the most suitable primary model**, while **DNN and Random Forest were retained as supporting models** for comparative analysis and ensemble-based future enhancements. This multi-model strategy strengthens system reliability and ensures resilience under diverse real-world GNSS operating conditions.

6. Plan for Next Week (Final Week)

The final week will focus on consolidating all experimental results, finalizing the internship report and research paper, preparing the final presentation, and validating the complete spoofing detection framework for demonstration and evaluation.

7. Learning Summary

This week significantly strengthened understanding of real-world GNSS security challenges, advanced machine learning model behavior, and the importance of robust evaluation using realistic datasets. The process of model selection reinforced the value of empirical performance over theoretical assumptions.

8. Verification

Intern Signature:



Date: August 24, 2025

Faculty Mentor Signature:



Date: August 24, 2025

WEEK .8 REPORT

1. Basic Information

Field	Details
Intern Name	YUNES ABDULGHANI MOHAMMED GHALEB
Enrollment No.	23BCAR0626
Institution	JAIN (Deemed-to-be University)
Faculty Mentor / PI	Dr. K. Suneetha, Dr. Zion Ramdinthara
Week No.	Week 8 (August24- Sep1, 2025)
Reporting Date	Sep1, 2025

2. Project Details

Project Title:

Project Title: Detection & Classification of Spoofing Attacks on GNSS Signals

Using AI-Based Models on NavIC System

Focus Area:

**Real GNSS/NavIC Data Analysis, Advanced Machine Learning Models, Model
Selection and Optimization**

2. Project Details

Project Title: Livestock Monitoring using LoRa and NavIC
Focus Area: Final Integration | Documentation | Presentation & Research Output
Objective for this Week:

This final week focused on **project consolidation and closure**, including comprehensive review of all technical work completed during the internship, preparation of final documentation, refinement of system outputs, preparation of presentation and research materials, and successful demonstration of the complete system to the faculty panel.

3. Tasks Undertaken (Final Review & Closure Phase)

During the final week, all technical, analytical, and documentation-related activities were systematically reviewed and finalized to ensure completeness, clarity, and professional quality.

A **comprehensive final internship report** was prepared, consolidating the entire eight-week work into a structured technical document covering system architecture, hardware integration, communication protocols, experimental results, machine learning evaluations, and future enhancement plans. Special attention was given to consistency, clarity of explanations, and alignment with academic reporting standards.

In parallel, a **final presentation deck** was designed to clearly communicate the project objectives, methodology, innovations, and results within a limited time frame. The slides were structured to include both a high-level executive summary and technical deep-dive sections, supported by diagrams, graphs, and experimental outcomes.

A **final live system demonstration** was conducted before the faculty evaluation panel. The demonstration showcased the complete end-to-end workflow—from sensor data acquisition and NavIC positioning, through LoRa transmission, to real-time visualization on the web dashboard—confirming the operational readiness of the system.

All **hardware and software modules** underwent final validation testing to confirm system stability, correctness, and integration integrity. Backup materials, including a

pre-recorded demonstration video, were also prepared to ensure robustness against unforeseen technical issues during evaluation.

Additionally, all **project files, datasets, source code, schematics, and experimental logs** were organized into a structured repository with clear documentation. A **Phase 2 roadmap** was formally documented, outlining future enhancements such as UWB integration, geofencing, cloud deployment, and advanced analytics.

4. Results / Outputs (Final Deliverables & Outcomes)

Final System Output

The project concluded with a **fully functional livestock monitoring system**, integrating:

- ESP32-based cattle units with LoRa communication
- NavIC-based positioning with real-time tracking
- Web-based dashboard for visualization and monitoring
- Local database for data storage and analysis

All components were validated together, demonstrating reliable operation under real-world conditions.

Documentation & Research Outputs

- A **comprehensive final internship report** documenting the complete technical journey
- Well-structured hardware schematics and firmware documentation
- Detailed explanation of experimental results, including field testing and model evaluations
- A **research-oriented analysis** discussing system limitations, trade-offs, and future directions
- A documented **UWB feasibility study**, positioning it as a Phase 2 enhancement

Presentation & Demonstration Materials

- Final presentation slides suitable for academic evaluation
- Primary and backup demonstration videos
- Live demonstration hardware setup

Project Closure Achievements

- Successful completion of all internship objectives
- Verified end-to-end system reliability
- Clear articulation of design decisions supported by experimental data
- Professional-quality documentation enabling reproducibility and future extension

5. Challenges and Solutions (Final Phase)

During the final phase, challenges primarily revolved around **time management, presentation clarity, and system reliability during demonstrations**. These were addressed through prioritization, preparation of backups, and iterative refinement of documentation and visuals.

6. Project Completion Summary

This week marked the **formal closure of Phase 1** of the project. All planned objectives were achieved, and the system was validated both technically and academically. The project now stands as a **complete, deployable prototype**, with a clearly defined roadmap for future expansion.

7. Learning Summary (Final Reflection)

The final week reinforced the importance of:

- Proper documentation and presentation of technical work
- Systematic review and validation before project closure
- Translating complex engineering work into clear academic and professional outputs
- Defending design choices using data-driven results

This phase significantly strengthened skills in **technical writing, system integration review, research articulation, and professional presentation.**

8. Verification

Intern Signature:



Date: Sep1, 2025

Faculty Mentor Signature:



Date: Sep1, 2025

Acknowledgments:

I am profoundly thankful to JAIN (Deemed-to-be University) and the IIT Tirupati Navavishkar I-Hub Foundation (IITTNiF) for providing me with the invaluable opportunity to undertake my internship under the Positioning, Navigation and Timing (PNT) Laboratory. This internship has been a truly enriching and rewarding experience that has significantly enhanced my technical knowledge, research exposure, and professional competencies.

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